The Cosmochrony Research Programme: A Structural Roadmap and Paper Inventory

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Abstract

The Cosmochrony programme develops a pre-geometric framework in which physical structure — spacetime, quantum mechanics, gauge symmetry, and Standard Model observables — arises from a single primitive: the local structure of admissible non-injective transitions between observable states. The corpus comprises 70 papers across three theory branches. Branch I (Foundation track) builds the axiomatic base: four axioms suffice to derive the Heisenberg group $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and its Weil representation as theorems, without any background substrate or relaxation mechanism. Branch II (O-series) develops the spectral admissibility sub-programme, deriving and numerically validating the transfer chain $c_\chi \to \delta_{\mathrm{pair}} \to \beta^* \approx 0.126$ connecting the BI saturation constant to the charged-lepton mass hierarchy. Branch III (Q-series and companion papers) derives quantum mechanics, spacetime geometry, gauge structure, and further physical observables from Branch I axioms and Branch II spectral data. Paper Q3 completes the SU(2) quantum sector: by combining BI indiscernibility with Schur's lemma on the Clebsch–Gordan decomposition of $\chi_{2j+1} \otimes \chi_{2j+1}$, the universal singlet correlator $E(\hat{a}, \hat{b}) = -j(j+1)/3 \cdot (\hat{a} \cdot \hat{b})$ and the Born rule are derived for all five admissible sectors of $2I$, closing the general-j extension that was open in Q2. The geometric emergence sub-programme (Q5a–Q11, U1, W1, H2) closes the effective spacetime identification: under the Q5a continuum-limit hypotheses, the lifting hypothesis [H-lift] is proved (Q9), asymptotic $\mathfrak{su}(2)$-isotropy $A_H \to 2$ is proved (Q10, U1), the admissibility weight convergence [H-w] is proved (W1), and the temporal coefficient is identified as $A_\tau = 2$ by cascade internalization and Casimir rigidity (Q11), yielding the fully determined effective Lorentzian co-metric $g^{\mu\nu} = 2\,\eta^{\mu\nu}$ with no free parameter. Q5a v2.0 establishes that [H1] is not needed in full generality. Q5a-O2 [62] proves spectral atomicity of the admissible sector (each pair spans three pure Fourier modes) and closes hypotheses [H-$\mathcal{E}$1] and [C] without Nash inequalities. Hypothesis [H2] (strong convergence of the rescaled Weil generators) is proved (H2 paper), closing all hypotheses of the Q5a large-q convergence theorem unconditionally on the admissible sector. O31 develops the structural framework for the SU(3) identification: the metaplectic intertwining of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ forces triplet co-admissibility unconditionally for the colour-adapted Cayley graph, and SU(3) is thereby identified as the symmetry group of the co-admissible colour triplet conditional on [H-color]. This paper provides a complete inventory, a logical dependency map, and a structured account of what is proved, structural, heuristic, or open. It is intended both as an entry point for external readers and as an internal navigation reference.

Keywords. Non-injective projection; admissibility; Born–Infeld constraint; spectral admissibility; Heisenberg group; Weil representation; emergent spacetime; lepton mass hierarchy; quantum mechanics; gauge structure; programme overview.

Contents

1 Introduction and Purpose

The Cosmochrony programme was initiated with a white paper [13] positing that all physical structure emerges from a non-injective projection $\Pi : \Omega \to \mathcal{O}$ from a pre-geometric substrate χ onto observable space. The initial framework postulated a Born–Infeld (BI) saturation constraint, a relaxation mechanism, and an iterated projection cascade $U_{t+1} = \Pi \circ \sigma(U_t)$ as the engine of dynamics.

Over the course of 2026, the programme has evolved in two parallel directions. On one side, the spectral admissibility sub-programme (O-series, papers O1–O28) has progressively built the computational machinery for extracting physical observables from the spectral structure of the substrate, culminating in the proof of the unconditional transfer chain in O24 [56]. On the other side, the Foundation paper [4] has refounded the entire framework from four axioms (A1–A4), dispensing with any explicit substrate χ and relaxation mechanism: these emerge as consequences of the axiomatic structure rather than being postulated. The Q-series then builds quantum mechanics, SU(2) symmetry, and Lorentzian spacetime on top of the Foundation axioms and O-series spectral data.

The corpus comprises 70 papers with complex mutual dependencies. A reader entering the programme faces two difficulties: it is not obvious where to start, and the distinction between a theorem, a structural argument, and an open conjecture is not always transparent from individual paper titles. This roadmap addresses both difficulties.

Conventions on result status. Throughout this paper each result is classified as one of the following: **proved** (formally proved as a theorem within the stated assumptions); **structural** (derived from structural arguments that are internally consistent but not yet fully formalised); **heuristic** (physically motivated and numerically supported but not yet rigorously derived); **open** (explicitly conjectured or known to be unsettled). These classifications follow the original papers; this roadmap introduces no new claims.

2 Conceptual Architecture

The programme is organised in three structural levels.

Level 1 — Axiomatic primitive. The primitive is the local structure of admissible non-injective transitions. No background spacetime, no quantum postulate, and no substrate field are introduced at this level. The four axioms are: local projective admissibility (A1), structural non-injectivity (A2), admissibility as non-premature selection (A3), and projection locking (A4). From A1–A2 alone, irreversibility and the arrow of time follow. From A3, the Heisenberg non-commutativity and the Weil representation emerge. From A4, the discreteness of quantum transitions follows. Papers at this level: ENI [28], Foundation [4], HeisenbergStructure, noscale [26].

Level 2 — Spectral fibre structure. The admissible fibre is proved to carry the Weil representation V_ρ of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for a prime q (Foundation Theorem 5.6; detailed proof in HeisenbergStructure [27]). The O-series computes the spectral properties of this fibre — in particular the capacity exponent δ_{pair} and its transfer to the cascade exponent β^* governing the charged-lepton mass hierarchy. Papers at this level: SpectralAdmissibility [46], SpectralCapacity [47], SpectralGramRigidity [31], SpectralRelaxation [6], and O1–O31 ([37]–[63]).

Level 3 — Physical observables. Effective spacetime geometry, gauge structure, Standard Model charges and masses, gravitational waves, and cosmological observables are derived as consequences of Levels 1–2. No additional physical postulates are introduced beyond the continuum-limit hypotheses of Q5a–Q5b. Papers at this level: the Q-series (Q1 [18], Q2 [40], Q3 [67], Q5a–Q7), the gravity track (Spectral2 [42], Gravity [21], Thermodynamics, Lorentz [11]), and the companion papers ([8, 10, 45, 12, 20, 55, 54]).

3 Branch I: Axiomatic Foundations

3.1 The White Paper

The white paper [13] ("Cosmochrony: A Non-Injective Projection Theory of Emergent Physics," v3.1) is the programme's primary overview document. It introduces the substrate field χ, the non-injective projection Π, the BI saturation constraint, the relaxation mechanism, and the qualitative derivation of spacetime and quantum structure. The spectral entropy functional $S_\Pi[g] = \frac{1}{2}\log\det' A_g$ appears here as the structural origin of the Einstein equation.

Role: Programme overview; structural motivation; first formulation of the BI constraint and the relaxation cascade. *Evolution note:* The explicit substrate χ and relaxation mechanism of the white paper are not postulated in the Foundation reformulation: they emerge from axioms A1–A4. The white paper remains the primary narrative reference; Branch I papers are the formal foundation.

3.2 Non-Injectivity as a Structural Necessity of Genuine Emergence

The ENI paper [28] proves the following no-go theorem. Any genuinely emergent (non-isomorphic effective) physical description is necessarily non-injective and intrinsically compressive. The argument is purely structural: given a surjective $\Pi : \Omega \to \mathcal{O}$ such that $\mathcal{O}$ carries no independent microstate-resolving structure beyond Π, non-isomorphism of levels forces non-injectivity. A projection entropy $S_\Pi \geq 0$ is defined, and the equivalence genuine emergence $\Leftrightarrow$ Π non-injective $\Leftrightarrow$ $S_\Pi > 0$ is established (**proved**, Theorem 2.1 and Corollary 3.1).

Role: Foundational justification for structural non-injectivity (Axiom A2). Cited by Foundation, HeisenbergStructure [27], Q1 [18], Bell.

3.3 Foundation: Admissible Non-Injective Transitions as the Primitive

The Foundation paper [4] establishes the minimal axiomatic basis of the programme. From the four axioms A1–A4 alone it derives: (i) irreversibility and the arrow of time (A1+A2); (ii) the proto-state as a physically real unresolved configuration retaining phase coherence (A3); (iii) a non-trivial commutator between conjugate generators, forced by A3 and the minimality of A1; (iv) the identification of the admissible fibre symmetry group as $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for a prime q, together with BI parity (A1–A3); (v) the identification $F_n \cong V_\rho$ (Weil representation) as a theorem (Theorem 5.6); (vi) the discrete character of quantum transitions from A4.

Status: **proved** for the algebraic identification of the fibre; **structural** for the continuum limit (deferred to Q5a–Q5b). *Role:* Axiomatic spine. Cited by all Q-series papers, HeisenbergStructure [27] , noscale, and recent companion papers.

3.4 Non-Factorisability and the Emergence of Heisenberg Structure

HeisenbergStructure [27] provides the detailed proof of Foundation Theorem 5.6, omitted from the Foundation paper for concision. The central structural lemma — *admissible non-factorisability* — is a direct formalisation of Axiom A3 and requires no input beyond A1–A3, BI parity, and classical group-classification results. As a corollary, the Heisenberg uncertainty principle is derived as a structural property of the non-resolved proto-state, not as an independent postulate (**proved**).

Role: Detailed proof of the algebraic core; cited by Q1 [18], Q2 [40].

3.5 No External Dimensional Scale at the Level of Admissibility

The noscale paper [26] establishes a structural rigidity result (Level 1): any deformation of the admissibility constraints that introduces a dimensional parameter independent of c_χ breaks at least one of BFS-consistent spectral scaling, structural non-injectivity (A2/ENI), or bounded admissibility closure (A3–A4). Dimensional analysis within the framework is therefore fully constrained by c_χ alone (**proved**).

Role: Closes the naturalness question for dimensional parameters.

4 Branch II: Spectral Admissibility Sub-Programme (O-Series)

The O-series targets the derivation of the charged-lepton mass hierarchy from the spectral structure of the admissible fibre. The central quantity is the power-law exponent δ_{pair} of the pair-level capacity observable $\sigma_{\text{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$ along the BFS cascade. The main transfer chain is

$$c_\chi \longrightarrow \delta_{\text{pair}} \longrightarrow \beta^* \approx 0.126 \longrightarrow m_e : m_\mu : m_\tau. \tag{1}$$

This chain is proved unconditionally with respect to the fibre structure of Π (O24 [56], Corollary 4.2). The numerical validation of (1) is the subject of O25 [51].

4.1 Precursor Papers: SU(2) Admissibility Sub-Programme

Three papers precede the O-series proper and explore spectral admissibility on Cayley graphs of finite SU(2) subgroups.

SpectralAdmissibility [46] (v2.0) carries out the spectral admissibility programme on two parallel chains. For the SU(2) chain $Q_8 \subset 2I \subset \mathrm{PSL}(2, \mathbb{F}_q)$, it derives the admissibility hierarchy $A_n^{\max} = c_\chi/\sqrt{\lambda_n}$ in the linearised regime and shows that the non-linear DBI correction preserves it. For the SU(3) chain $\Delta(27) \subset \Sigma(168)$, it establishes via character theory the hierarchy $\lambda_{\mathbf{3}} = \lambda_{\bar{\mathbf{3}}} = 28 < \lambda_{\mathbf{6}} = \lambda_{\mathbf{8}} = 42 < \lambda_{\mathbf{7}} = 48$ on $\mathrm{Cay}(\Sigma(168), C_4)$, proving that the fundamental and anti-fundamental representations are co-admissible and strictly dominate all competitors. The algebraic mechanism is uniform: $\chi_{\mathbf{3}}(g) = 1$ for all $g \in C_4$ plays the same structural role as $\chi_{\rho_4} = 0$ on Q_8 and the golden-ratio identity on $2I$ (**proved**).

SpectralCapacity [47] introduces the capacity functional $\mathcal{C}(G, S)$ and proves the binary-polyhedral maximality conjecture: the adjoint identity $\hat{A}_{\text{adj}} = 2M - I$ (where M is the second-moment tensor of the axis distribution) shows that icosahedral isotropy forces $\mu_{\text{adj}}(2I) = -\frac{1}{3}$, while axis-plane confinement forces $\mu_{\text{adj}}(\mathrm{Dic}_n) \geq 0$ for all $n \geq 3$; combined with a Plancherel uniform bound, this yields a strictly increasing capacity gap $\Delta(\alpha) > 0$ for all $\alpha > 0$ (**proved**).

SpectralGramRigidity [31] (v1.0.1) shows that the geometry of neutral generating sets on SU(2) is fully encoded in pairwise character values $G_{st} = -\frac{1}{2}\chi_\rho(st)$ (**proved**).

Three Particle Generations [52] derives the three-generation structure from the ADE stratigraphic levels of binary Cayley graphs (**structural**).

4.2 SpectralRelaxation: Expander Graph Dynamics

The SpectralRelaxation paper [6] establishes $\Lambda_{\text{proj}} \asymp \lambda_2$ for LPS expander families (**proved**), providing the entry point for the O-series dynamics.

4.3 O1–O8: LPS Phase — Cascade Exponent β

O1 [37]: LPS saturation is a fixed-valence artefact; growing valence restores the pre-saturation window (**proved**). (Note: O2 was not written; the O3 derivation path was chosen instead.)

O3 [35]: First quantitative SM connection. The phenomenological window $\beta^* \in (0.09, 0.13)$ is derived from the charged-lepton mass ratios $m_e : m_\mu : m_\tau$ (**structural**).

O4 [50]: The structural upper bound $\beta \leq 1$ follows from the BI flux constraint and the Cheeger isoperimetric inequality (**proved**).

O5 [3]: Obstruction theorem. The β gap cannot be closed by any purely representation-theoretic or fixed-representation approach (**proved**).

O6 [24]: Matrix-level confinement of the cascade exponent (**proved**).

O7 [33]: Reformulates the target as a capacity exponent $\delta \in [7.4, 10.6]$ on three-step path fingerprints (**structural**).

O8 [1]: Establishes the geometric obstruction closing the LPS phase. Exponential shell growth $|S_n| \sim p^n$ on LPS graphs compresses the pre-saturation window to $O(\log q)$ BFS steps (**proved**).

4.4 O9–O15: Heisenberg Transition

The geometric obstruction identified in O8 motivates replacing LPS graphs by Cayley graphs of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$.

O9 [34]: Structural transition. Polynomial ball growth $|B_n| \sim n^4$ provides $\Theta(q)$ usable BFS steps (**proved**). This is the key structural transition of the O-series.

O10 [32]: First large-q computation ($q = 101$). Confirms n^4 growth; a TensorSketch representation obstruction is identified at large q (**proved**).

O11 [57]: Bidimensional Fourier-character proxy overcomes the TensorSketch obstruction (**structural**).

O12 [15]: Exact Weil-block fingerprints resolve the final obstruction. Establishes $\hat{\delta}_{\mathrm{exact}} \approx 4.3$–$4.8$ for $q \in \{29, 61, 101, 151, 211\}$ (**proved**).

O13 [7]: $\hat{\delta}_{\mathrm{exact}}(q)$ decreases monotonically from 4.80 to 3.59; the finite-size hypothesis is definitively refuted (**proved**).

O14 [30]: Corrects the $\delta \mapsto \beta^*$ relation via block normalisation and central-phase contributions (**structural**).

O15 [29]: Block-level no-go. No reweighting of individual block data can raise $\hat{\delta}_{\mathrm{exact}}$ to $\delta \in [7.4, 10.6]$ (Corollary 4.8, **proved**). This result forces the pair observable introduced in O16.

4.5 O16–O24: Pair Observable and Unconditional Transfer Chain

O16 [16]: Identifies the misidentification of the observable. The physically relevant quantity is $\sigma_{\mathrm{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$ on conjugate pairs $\{c, q-c\}$. The exponent doubling $\delta_{\mathrm{pair}} = 2\delta_c \approx 7.44$ resolves the gap (**structural**).

O17 [17]: Structural derivation. $\rho_{q-c} = \bar{\rho}_c$ is forced by fibre-level admissibility; the pair structure is derived, not postulated (**proved**).

O18 [25]: Foundational derivation of the parity involution $c \leftrightarrow q - c$. BI parity is the only global symmetry of $S_{\mathrm{BI}}[\chi]$, ensuring each fibre of Π is exactly $\{\chi, -\chi\}$ (minimal fibre condition, **proved**, Theorem 4.1).

O19 [2]: The residual amplitude factor $r(c, q) \in \{1, 2, 3, 4\}$ does not affect δ_{pair}; the observable is pipeline-independent (**proved**).

O20 [36]: Dynamical selection criterion for admissible modes via projective persistence (**structural**).

O21 [44]: Defines the observable saturation rank n_3^{obs} via $\Sigma_c(n_3) = 3$ and constructs the transfer $\delta_{\mathrm{pair}} \to \beta^*$ (**structural**).

O22 [43]: BI projection locking forces n_3^{obs} onto a BFS shell (Theorem 3.2, **proved**).

O23 [53]: The threshold $\Sigma_c(n_3) = 3$ follows from the quaternionic maximality of the admissible neutral generating set (**proved**).

O24 [56]: Capstone of the analytical O-series. The Verticality Lemma ("$\ker \Pi$ is free, $\mathrm{Im}\, \Pi$ is constrained") closes the unconditional transfer chain $c_\chi \to \delta_{\mathrm{pair}} \to \beta^*$ (Corollary 4.2: $\beta^* \approx 0.126$, **proved**).

4.6 O25–O30 and PTO: Numerical Validation, Sector Identification, and Eigenvalue Ratio

O25 [51]: Systematic pair-level numerical campaign. Computes δ_{pair} across all $(q-1)/2$ conjugate pairs with $M = 50$ independent block samples per pair, for $q \in \{29, 61, 101, 151, 211\}$. Results: $\bar{\delta}_{\text{pair}} \in \{8.89, 9.98, 9.55, 9.13, 8.78\}$ with inter-pair standard deviation decreasing from 0.54 to 0.11, confirming δ_{pair} as a structural invariant. After the O14 normalisation correction, corrected values $\delta_{\text{corr}}(q) \in [7.7, 9.0]$ fall consistently within the admissible window $[7.4, 10.6]$ for $q \in \{61, 101, 151, 211\}$. Direct extrapolation to $q \to \infty$ is shown to be structurally degenerate; the central open direction is the analytical control of $n_1(q)/q$ (**structural**).

O26 [39]: Dictionary between conjugate Weil blocks and rank-1 matrices in $M_2(\mathbb{C})$, connecting the Heisenberg fibre to $2I \subset \mathrm{SU}(2)$ (**structural**).

O27 [41]: Every admissible morphism $\Phi_{q,\rho}$ satisfying involution equivariance, quadratic compatibility, and naturality factors through $\mathfrak{su}(2)$ (**proved**).

O28 [5]: Two measurements from the Q5a-O5 checkpoints. Part A: the BFS window ratio $n_1(q)/q$ decreases from 0.138 to 0.066 over $q \in \{29, 61, 101, 151, 211\}$ (Scenario S2, pre-asymptotic regime); OLS fit $n_1(q) = \hat{\alpha}\, q + \hat{\beta}$ gives $\hat{\alpha} \approx 0.053$; $\delta_{\text{corr}}(q) \in [7.4, 10.6]$ for all tested primes, confirming O25. Part B: formal effective-dimension computation of O26 Criterion 5.4 in $\mathrm{End}(H_{\text{eff}})$, $H_{\text{eff}} = \mathbb{C}^3$ (HEFF_DIM = 3, consistent with $\Sigma_c(n_3) = 3$ from O23). The covariance $\mathcal{C}_c$ has rank exactly 3 for every conjugate pair and every $q \in \{61, 101, 151\}$ (100% of pairs), with invariant eigenvalue structure $[\lambda_1 : \lambda_2 : \lambda_3] = [1 : \frac{1}{2} : \frac{1}{2}]$. The exact degeneracy $\lambda_2 = \lambda_3$ is a structural consequence of the Born–Infeld parity involution $c \leftrightarrow q - c$ (O18), which correlates conjugate projections and constrains $\mathcal{C}_c$ to a symmetric subspace. The gap from $d_\rho^2 = 4$ reflects $H_{\text{eff}} \neq V_\rho$; analysis of $V_\rho \subset H_{\text{eff}}$ is carried out in O29.

O29 [48]: Spin-$\frac{1}{2}$ sector identification via the symmetric rank formula (**structural**, numerically confirmed). The Born–Infeld parity constraint $\pi_{q-c}(v) = \overline{\pi_c(v)}$ (O18) is proved to force every outer product $M_j = \pi_c(v_j) \otimes \pi_{q-c}(v_j)^*$ into the complex symmetric subspace $\mathrm{Sym}(V_\rho, \mathbb{C}) \subset \mathrm{End}(H_{\text{eff}})$. This is an algebraic identity, not lifted by any complex-linear reparametrisation. The resulting covariance rank satisfies the *symmetric rank formula* $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$, whose unique positive integer inversion at $r_{\text{eff}} = 3$ gives $d_\rho = 2$, confirming the spin-$\frac{1}{2}$ sector. Numerically: $r_{\text{eff}} = 3$ universally for $q = 61$ (30/30 pairs) and $q = 151$ (75/75 pairs) with zero inter-pair variance; at $q = 211$ (103/105 pairs, 98%), $r_{\text{eff}} = 3$ holds for all but two pairs, consistent with a single-sample Monte Carlo artefact; at $q = 101$, modal $r_{\text{eff}} = 3$ is confirmed by a dedicated multi-sample analysis ($M = 5$ resamples per anomalous pair, 89% of cells satisfy $r_{\text{eff}} = 3$), resolving the single-sample anomaly as a Monte Carlo artefact. The O26 Criterion 5.4 is reformulated for conjugate-pair data: $d_\rho^2 = 4 \to d_\rho(d_\rho + 1)/2$; spin-$\frac{1}{2}$ satisfies the reformulated criterion exactly.

O30 [59]: Analytical derivation of the eigenvalue ratio $\lambda_1/\lambda_2 = 2$ (**proved**). The Born–Infeld parity anti-linearity $\rho_{q-c} = \bar{\rho}_c$ forces every admissible trajectory vector $w_j = \pi_c(v_j) \in V_\rho \cong \mathbb{C}^2$ to satisfy $|\alpha_j| = |\beta_j|$ (equatorial constraint), i.e., w_j lives on an equatorial $S^1 \subset S^3$ in V_ρ. In the equatorial regime, the central component $\sqrt{2}\,\alpha_j\beta_j$ of $\mathrm{vec}(M_j)$ is phase-independent (the phases cancel exactly), while the diagonal components α_j^2 and β_j^2 oscillate as $e^{\pm 2i\phi_j}$. The covariance then diagonalises under the single-harmonic phase decorrelation condition $\langle e^{2i\phi_j} \rangle = 0$, yielding $\mathcal{C}_c = \langle r_j^4 \rangle \cdot \mathrm{diag}(1/4, 1/2, 1/4)$ and spectral ratio $[1 : \frac{1}{2} : \frac{1}{2}]$ exactly. The factor of 2 is purely structural: it is the squared symmetric-tensor normalisation coefficient in $e_0 = (v_+ \otimes v_- + v_- \otimes v_+)/\sqrt{2}$, isolated by the equatorial constraint as the sole source of eigenvalue asymmetry. The correction to the equatorial condition is $O(q^{-1/2})$ from U1 [66]. This closes the open direction of O28–O29 §7.4.

O31 [63]: Structural framework for the SU(3) identification (**structural**). Shows the O23 Hurwitz argument cannot directly produce $d_\rho = 3$ (proved). Defines the *colour triplet* $\{c_1, c_2, c_3\} \subset (\mathbb{Z}/q\mathbb{Z})^*$ with $c_1 + c_2 + c_3 \equiv 0 \pmod{q}$; such triplets exist iff $q \equiv 1 \pmod{3}$ (proved). The metaplectic automorphism $\phi_\omega(a, b, z) = (\omega a, \omega^{-1} b, z)$ forces $\sigma_c(n) = \sigma_{\omega c}(n)$ for all shells on the colour-adapted Cayley graph $\mathrm{Cay}(G_q, \mathcal{S}_q^{(3)})$ (proved, Theorem 4.4). Conditional on Hypothesis [H-color] (triplet co-admissibility for the standard graph), the symmetry group of the co-admissible triplet is exactly SU(3), with U(3), SO(3), U(2), G_2 excluded (proved). The Standard Model gauge group $G_{\mathrm{SM}} = \mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ follows as a composite of three admissibility fixed points (proved for $\mathcal{S}_q^{(3)}$; conditional on [H-color] for $\mathcal{S}_q$). U1 universality implies $|\sigma_c - \sigma_{\omega c}| = O(q^{-1/2})$ for any generating set, so the identification is generating-set-independent in the large-q limit. Numerical test deferred to O32 ($q \in \{61, 151, 211, 307\}$, $q \equiv 1 \pmod 3$).

PTO [38]: The cumulative projected capacity $I(n) = \sigma_{\mathrm{pair}}(0) - \sigma_{\mathrm{pair}}(n)$ is strictly non-decreasing and hence compatible with an arrow of time. Confirmed numerically for $q \in \{29, 61, 101, 151\}$ (**structural**).

5 Branch III: Q-Series and Physical Applications

5.1 Q-Series: Quantum Mechanics and Spacetime Emergence

The Q-series derives quantum-mechanical and geometric structure from the Foundation axioms, taking as input the fibre identification $F_n \cong V_\rho$ and selected O-series results.

Q1 [18]: Phase coherence is a structural consequence of projective admissibility (not a postulate). Theorem 2.7 proves that any admissible transition preserves the BI indiscernibility of conjugate Weil blocks ρ_c and ρ_{q-c}, maintaining the metaplectic phase coherence of the fibre. The singlet correlator $E(\hat{a}, \hat{b}) = -\hat{a}\cdot\hat{b}$ and the Tsirelson bound are derived from admissibility, O18, and O23 (**proved**; numerically validated for $q = 29$).

Q2 [40]: Extends the programme to spin-$\frac{3}{2}$. On the binary icosahedral group $2I$, the spin-$\frac{1}{2}$ and spin-$\frac{3}{2}$ sectors share the Laplacian eigenvalue $\lambda_{1/2} = \lambda_{3/2} = 18$ (co-admissibility). The Born rule, singlet correlator, and Tsirelson bound are derived for spin-$\frac{3}{2}$ without new postulates. SU(2) is established as the stable fixed point of the admissibility flow (**proved** for spin-$\frac{3}{2}$; general-j extension proved in Q3 [67]).

Q3 [67]: Completes the SU(2) quantum sector for all admissible j. BI indiscernibility forces the bipartite proto-state to be invariant under the diagonal $2I$-action on $V_{\chi_{2j+1}} \otimes V_{\chi_{2j+1}}$; Schur's lemma applied to the Clebsch–Gordan decomposition $\chi_{2j+1}\otimes\chi_{2j+1} = \bigoplus_{J=0}^{2j}\chi_{2J+1}$ uniquely identifies the proto-state as the singlet $|\Omega_j\rangle$. The universal correlator $E(\hat{a}, \hat{b}) = -j(j+1)/3\cdot(\hat{a}\cdot\hat{b})$ and the Born rule are derived for all five admissible sectors $j \in \{\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}\}$ of $2I$ (**proved**). Closes the open step of Q2 (identification of the isotropic fibre state with the proto-state for spin-$\frac{3}{2}$, and general-j extension).

Q5a [22]: Under Mosco hypotheses, the admissibility forms $\mathcal{E}_q$ on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ converge to $L_\Pi = -A\partial_x^2$ on $L^2(\mathbb{R})$ (**structural**, conditional).

Q5b [9]: The BFS shell stratification of $\mathrm{Heis}_3(\mathbb{R})$ produces a four-dimensional Lorentzian spacetime, with the temporal direction identified with the BFS depth. Hypothesis [H-lift] (Q5b-O1), formerly conditional, is proved by Q9 [60] (**structural**; unconditional with respect to [H-lift] via Q9).

Q6a [19]: Gauge charges are the invariants of the projection-fibre equivalence classes. U(1) and SU(2) are identified as fixed points of the admissibility flow. $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ is derived as a composite (**structural**; SU(3): conditional on [H-color], see O31 [63]).

Q6b [14]: Effective Lorentzian geometry from the admissibility filter Π_q; inherits Q5a hypotheses only ([H-lift] closed by Q9) (**structural**, conditional on Q5a).

Q7 [65]: Bridge candidate between the admissible dimension $\dim H_{\mathrm{eff}} = 3$ (O23–O29) and the three spatial dimensions of Q5b. Proves $[\mathfrak{su}(2)] \ncong \mathfrak{heis}_3$ (the bridge cannot be a Lie algebra isomorphism) and identifies $H_{\mathrm{eff}} \cong \mathrm{Sym}^2(V_\rho)$ as the spin-1 representation of su(2) (O29). Establishes structural uniqueness of the bridge via Schur's lemma, and reduces the identification to the isotropy condition $A_H = A_z$ on $\sigma_2(L_{\mathrm{eff}})$. Proves analytically $[F_c, L_{\mathrm{Weil}}] = 0$, which forces the cross terms $k_X k_Z = k_Y k_Z = 0$ structurally. Numerical evidence on O25 checkpoints ($q \in \{61, 101, 151, 211\}$) gives $A_z = 2$ universally and $|A_H - A_z| \to 0$; four-point OLS fit gives

$|A_H - A_z| \approx 0.364\, q^{-0.52}$ ($R^2 = 0.98$), consistent with $O(q^{-1/2})$ (Q5b Theorem 3.2). (**structural**; asymptotic isotropy $A_H \to 2 = A_Z$ proved by Q10 and U1, conditionally on [U]; bridge existence at finite q: **open**).

Q8–Q11, U1, W1 (Geometric emergence cluster). Q8 [49] establishes $A_Z = C_{\mathfrak{su}(2)} = 2$ unconditionally under Q5a via Casimir rigidity (**structural**). Q9 [60] proves [H-lift] as a theorem: the modulation generator is suppressed at rate $O(q^{-1/2})$, and $A_H > 0$ follows by coercivity independently of the bridge (**proved**, under Q5a). Q10 [58] proves $A_H \to 2$ under spectral universality [U]: character-independence forces $\mathfrak{su}(2)$-isotropy of the effective form, and the unique such form is the Casimir with value 2 (**structural**, conditional on [U]). U1 [66] proves [U] with rate $\varepsilon(q) = O(q^{-1/2})$ from the Weil-generator Lipschitz bound and BFS–Carnot–Carathéodory convergence (**proved**). W1 [68] proves Hypothesis [H-w] (convergence $a_q(s) \to A > 0$ of the admissibility weights) as a structural consequence of the fibre-invariance of admissible observables ($\mathcal{O} = \operatorname{Im}\Pi$), made quantitative by U1 (**proved**). Q11 [64] closes open problem Q5b-O3 by proving $A_\tau = 2$ (**structural**, conditional on [U]): the temporal coordinate τ is identified with the BFS depth index n (Q5b Remark 3.3), the cascade increment operator T passes to the continuum derivative ∂_τ by Carnot–Carathéodory convergence, [U] forces T to be asymptotically SU(2)-equivariant, and Schur's lemma with the Q8 normalisation gives $A_\tau = 2$. The effective co-metric is thereby uniquely determined: $g^{\mu\nu} = 2\,\eta^{\mu\nu}$, with no remaining free parameter. After W1, the open Q5a hypotheses reduce to three: [H1], [H-E1], and [C]. Q5a-O2 [62] subsequently proves that [H1] is not needed in full generality on the admissible sector, and closes [H-E1] and [C] via spectral atomicity. H2 [61] proves the last open hypothesis, [H2] (strong convergence of the rescaled Weil generators $\hat{X}_q \to x$ and $\hat{P}_q \to -i\partial_x$), via a quantitative Poisson aliasing lemma and a discrete Sobolev identity (**proved**). All hypotheses of the Q5a large-q convergence theorem are verified unconditionally on the admissible sector.

5.2 Gravity and Spacetime Track

Spectral2 [42]: Derives operational metric distances from graph Laplacians on a relational substrate without any background manifold (**structural**).

Gravity [21]: Derives $G_{\mu\nu} = 8\pi G_N T^{(\Pi)}_{\mu\nu}$ from the metric variation of $S_\Pi[g] = \frac{1}{2}\log\det' A_g$, via heat-kernel and zeta regularisation (**proved** at leading IR order).

Thermodynamics [23]: Develops a local thermodynamic interpretation. The diagonal heat kernel yields a local multiplier field $\beta(x)^{-1} = \beta_0^{-1} - \frac{1}{6}R(x) + O(\nabla^2 R)$, and a spectral first law $\delta S_\Pi = \int \beta(x)^{-1}\,\delta E_\Pi(x)$ recasts the Einstein equation as a spectral equilibrium condition (**structural**; Lorentzian extension and matter coupling: **open**).

Lorentz [11]: Lorentzian completion of the spectral action. Causal propagation and gravitational wave propagation follow from the hyperbolic continuation (**proved** for the Lorentzian extension; GW spectrum: **structural**).

5.3 Standard Model Companion Papers

BornInfeld [10]: BI action as the unique minimal local representation consistent with a universal bound on admissible relational gradients (**proved**).

TopologicalInvariants [54]: Charge quantisation ($\pi_1(S^1) \cong \mathbb{Z}$) and absence of magnetic monopoles from the Hopf fibration $S^1 \hookrightarrow S^3 \to S^2$ of the projection fibre (**proved**; $\pi_2(\mathcal{C}_{\text{eff}}) = 0$ beyond the canonical model: **open**, Remark 4.2).

SMchargemass [12]: Mass and electric charge as BI-saturated responses (**structural**; baryon-to-photon ratio η: **open**).

Bell [8]: Bell-inequality violations from non-injectivity alone, without nonlocality or additional hidden variables (**proved**).

SchwingerLamb [20]: Fine-structure constant α as a projective invariant; Lamb shift and Schwinger correction rederived in the pre-geometric framework (**structural**; quantitative radiative corrections: **heuristic**).

5.4 Cosmological and Condensed-Matter Applications

Cosmology [45]: Flat galaxy rotation curves and the Hubble tension from a common effective mechanism; no dark matter or modified gravity (**structural**; quantitative fit: **heuristic**).

Superconductivity [55]: Projective phase locking as the structural origin of superconducting coherence across BCS, cuprate, and nickelate systems (**structural**).

6 Dependency Structure

Figure 1 presents the logical dependency DAG. Three entry paths emerge for external readers.

Foundational path. ENI → Foundation → HeisenbergStructure → noscale. Requires least prior knowledge; establishes the axiomatic core.

Spectral path. BornInfeld → SpAdm → SpRel → O1 → ⋯ → O24 → O25 → O28 → O29. Develops the spectral machinery and the transfer chain.

Physical path. Foundation + O18 + O22 + O23 → Q1 → Q2 → Q3; Q2 → Q5a → Q5b → Q6a + Q6b; O23 + O28 + O29 + Q5b → Q7 (dimensional bridge); parallel: Spectral2 → Gravity → Thermodynamics → Lorentz.

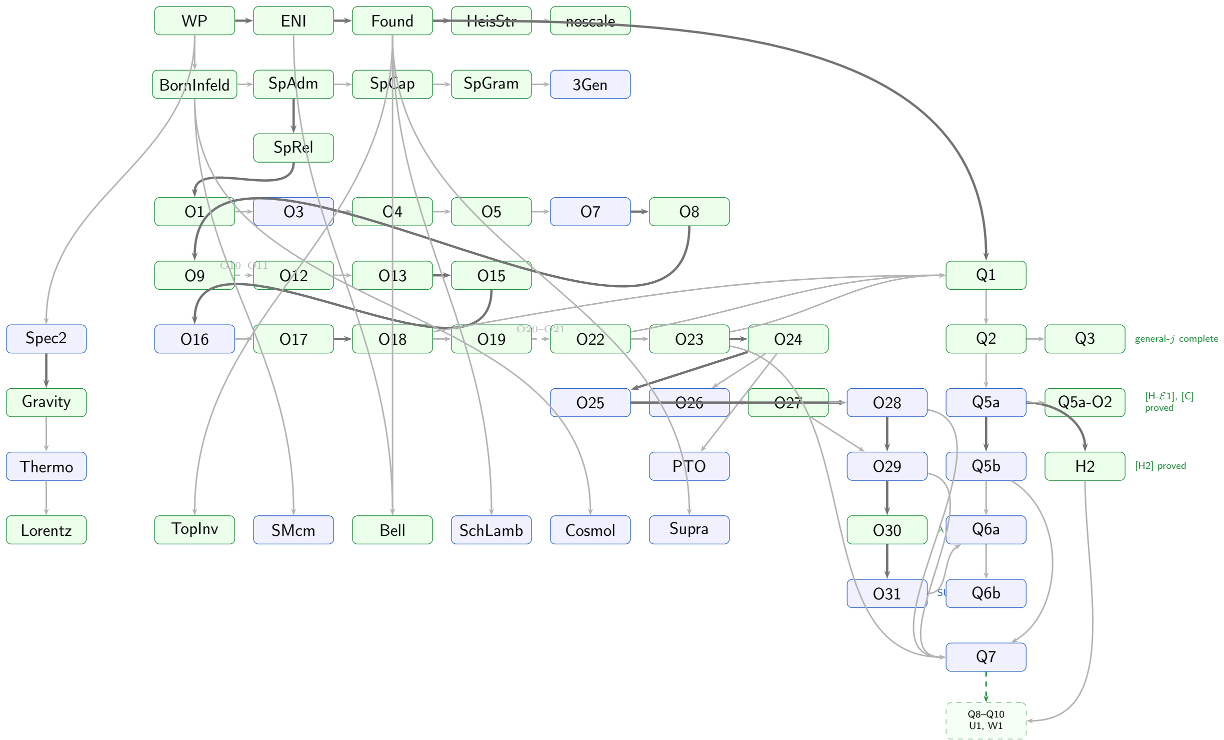

Figure 1: Logical dependency DAG of the Cosmochrony programme. Colour coding: ■ **proved**, ■ **structural**, ■ **heuristic**, ■ **open**. Thick arrows indicate the main logical spine; thin arrows indicate supporting dependencies. Dashed arrows with labels abbreviate linear sub-chains. The dashed green arrow from Q7 leads to the geometric emergence closure block Q8–Q10 and U1, detailed in Figure 2. The interactive version with clickable nodes is provided as a companion HTML file.

Table 1 lists all papers in the programme. The **Key** column gives the `cosmochrony-bibliography` citation key.

Label	**Short title**	**Key**	**Main contribution**	**Status**
WP	White Paper	[13]	Programme overview; substrate χ; iterated projection; BI constraint; relaxation cascade	structural
ENI	Emergence & non-injectivity	[28]	No-go: genuine emergence $\Leftrightarrow$ Π non-injective; projection entropy S_Π	proved
Found	Foundation	[4]	Axioms A1–A4; derives $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and V_ρ as theorems (Thm 5.6)	proved
HeisStr	Heisenberg structure	[27]	Detailed proof of Found. Thm 5.6; uncertainty principle from A3 alone	proved
noscale	No external scale	[26]	No independent dimensional parameter beyond c_χ	proved
BornInfeld	Born–Infeld geometry	[10]	BI action as unique minimal bounded-flux representation	proved
SpAdm	Spectral admissibility (Q_8)	[46]	$A_n^{\max} = c_\chi/\sqrt{\lambda_n}$; first tractable instance	proved
SpCap	Spectral capacity	[47]	Capacity functional $\mathcal{C}(G,S)$; binary-polyhedral maximality proved via adjoint identity $\hat{A}_{\mathrm{adj}} = 2M - I$ and icosahedral isotropy	proved
SpGram	Gram rigidity (v1.0.1)	[31]	Neutral-generator geometry from $G_{st} = -\frac{1}{2}\chi_\rho(st)$	proved
3Gen	Three generations	[52]	Three-generation structure from ADE stratigraphic levels	structural
SpRel	Spectral relaxation	[6]	$\Lambda_{\mathrm{proj}} \asymp \lambda_2$ for LPS expander families	proved
O1	Variable-valence graphs	[37]	LPS saturation is a fixed-valence artefact; growing valence restores window	proved
O3	Mass hierarchy	[35]	First SM connection: $\beta^* \in (0.09, 0.13)$ from lepton mass ratios	structural
O4	Upper bound $\beta \le 1$	[50]	BI flux + Cheeger $\Rightarrow \beta \le 1$	proved
O5	Rep-theoretic obstruction	[3]	Fixed-repr. approaches cannot close the β gap	proved
O6	Matrix redundancy	[24]	Matrix-level confinement of cascade exponent	proved
O7	Continuum target	[33]	Target: $\delta \in [7.4, 10.6]$ on 3-step fingerprints	structural
O8	LPS geometric obstruction	[1]	Exponential shell growth $\Rightarrow$ usable window $= O(\log q)$ BFS steps	proved
O9	Heisenberg graphs	[34]	Structural transition: $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ with $\lvert B_n\rvert \sim n^4$ replaces LPS	proved
O10	Large-q computation	[32]	Confirms n^4 growth; TensorSketch obstruction identified	proved
O11	Fourier-character proxy	[57]	Bidimensional proxy overcomes TensorSketch obstruction	structural

continued...

Label	Short title	Key	Main contribution	Status
O12	Exact Weil blocks	[15]	$\hat{\delta}_{\mathrm{exact}} \approx 4.3$–$4.8$ for $q \leq 211$; final obstruction resolved	proved
O13	Extended prime range	[7]	$\hat{\delta}(q) : 4.80 \to 3.59$; finite-size hypothesis refuted	proved
O14	Observable-class correction	[30]	Corrects $\delta \mapsto \beta^*$ via block normalisation	structural
O15	Block-level no-go	[29]	No reweighting of individual blocks reaches $\delta \in [7.4, 10.6]$ (Cor. 4.8)	proved
O16	Pair observable	[16]	$\sigma_{\mathrm{pair}} = \sigma_c\sigma_{q-c}$; exponent doubling $\delta_{\mathrm{pair}} \approx 7.44$	structural
O17	Structural derivation of pair	[17]	$\rho_{q-c} = \bar{\rho}_c$ forced by fibre admissibility: pair is derived, not postulated	proved
O18	BI parity involution	[25]	Minimal fibre condition $\{\chi, -\chi\}$; BI parity is the unique global symmetry of S_{BI}	proved
O19	Canonical normalisation	[2]	$r(c, q)$ does not affect δ_{pair}; observable pipeline-independent	proved
O20	Persistence criterion	[36]	Dynamical selection of admissible modes via projective persistence	structural
O21	Shell saturation	[44]	Observable rank n_3^{obs} via $\Sigma_c(n_3) = 3$; transfer $\delta_{\mathrm{pair}} \to \beta^*$	structural
O22	Projection locking	[43]	BI projection locking forces n_3^{obs} onto a BFS shell (Thm 3.2)	proved
O23	Quaternionic minimality	[53]	$\Sigma_c(n_3) = 3$ from quaternionic maximality of neutral generators	proved
O24	Observable rank rigidity	[56]	Verticality Lemma; unconditional transfer $c_\chi \to \delta_{\mathrm{pair}} \to \beta^* \approx 0.126$ (Cor. 4.2)	proved
O25	Numerical campaign	[51]	$\bar{\delta}_{\mathrm{pair}} \in \{8.89, 9.98, 9.55, 9.13, 8.78\}$ for $q \in \{29, 61, 101, 151, 211\}$; inter-pair $\sigma_q : 0.54 \to 0.11$; $\delta_{\mathrm{corr}} \in [7.7, 9.0] \subset [7.4, 10.6]$	structural
O26	Quadratic completion	[39]	Dictionary: Weil blocks $\leftrightarrow$ rank-1 matrices in $M_2(\mathbb{C})$; connects Heis_3 to $2I \subset \mathrm{SU}(2)$	structural
O27	Quaternionic rigidity	[41]	Every admissible $\Phi_{q,\rho}$ satisfying three structural constraints factors through $\mathfrak{su}(2)$	proved
O28	Window calibration & r_{eff}	[5]	$n_1(q)/q$ decreasing (Scenario S2); $\delta_{\mathrm{corr}} \in [7.4, 10.6]$ confirmed; $r_{\mathrm{eff}} = 3$ in $\mathrm{End}(H_{\mathrm{eff}})$, eigenstructure $[1 : \frac{1}{2} : \frac{1}{2}]$ invariant across all pairs and $q \in \{61, 101, 151\}$	structural

continued...

Label	Short title	Key	Main contribution	Status
O29	Symmetric rank formula & sector ID	[48]	BI parity forces $M_j \in \mathrm{Sym}(V_\rho, \mathbb{C})$; $r_{\mathrm{eff}} = d_\rho(d_\rho+1)/2$; unique inversion at $r_{\mathrm{eff}} = 3$ gives $d_\rho = 2$ (spin-$\frac{1}{2}$); confirmed for $q \in \{61, 101, 151, 211\}$; $q = 101$ anomaly resolved by multi-sample analysis	structural
O30	Eigenvalue ratio λ_1/λ_2	[59]	BI parity forces equatorial constraint $\lvert\alpha_j\rvert = \lvert\beta_j\rvert$; phase-independent central component gives $\mathcal{C}_c = \langle r^4\rangle\,\mathrm{diag}(1/4, 1/2, 1/4)$; ratio $[1 : \frac{1}{2} : \frac{1}{2}]$ derived analytically; factor of 2 is purely structural (Carnot degree-2 of $Z = [X, Y]$)	proved
O31	SU(3) from colour triplets	[63]	Hurwitz obstruction ($d_\rho = 3$ cannot arise from single-fibre argument, proved); colour triplets exist iff $q \equiv 1 \pmod 3$ (proved); metaplectic intertwining proves $\sigma_c = \sigma_{\omega c}$ on $\mathcal{S}_q^{(3)}$ (proved, Thm 4.4); conditional on [H-color], SU(3) is the unique triplet symmetry group (proved; U(3), SO(3), U(2), G_2 excluded); G_{SM} derived for $\mathcal{S}_q^{(3)}$; O32 tests [H-color] on standard graph	structural
PTO	Temporal ordering	[38]	$I(n) = \sigma_{\mathrm{pair}}(0) - \sigma_{\mathrm{pair}}(n)$ is a non-decreasing temporal proxy; confirmed for $q \in \{29, 61, 101, 151\}$	structural
Q1	Phase coherence	[18]	Phase coherence from admissibility (Thm 2.7); singlet correlator and Tsirelson bound derived	proved
Q2	Beyond spin-$\frac{1}{2}$	[40]	Spin-$\frac{3}{2}$ quantum structure; SU(2) as admissibility fixed point	proved
Q3	Universal spin-j singlet	[67]	Schur + BI indiscernibility identify proto-state as singlet $\lvert\Omega_j\rangle$ for all $j \in \{\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}\}$; universal correlator and Born rule derived; closes Q2 open step	proved
Q5a	Large-q operator limit	[22]	Mosco convergence $\mathcal{E}_q \to L_\Pi = -A\partial_x^2$; conditional on [H2]. v2.0: [H1] established on admissible sector (not needed in full generality); [H-$\mathcal{E}$1] and [C] proved by Q5a-O2 [62]; [H2] proved by H2 paper: all hypotheses verified on the admissible sector	structural

continued...

Label	Short title	Key	Main contribution	Status
Q5a-O2	Spectral atomicity & Mosco closure	[62]	Admissible sector spectrally atomic: $\mathrm{ran}(\Pi_q^{(c)}) = \mathrm{span}\{e_0, e_{\xi_c}, e_{q-\xi_c}\}$, $\xi_c/q \to \omega_c \neq 0$. Proves [H-$\mathcal{E}$1'] (scaled coercivity, $\beta = 1$ optimal) and [C] (Mosco compactness via Bolzano–Weierstrass in $\mathbb{C}^2$, no Nash inequality needed). Numerical confirmation $K_q = 1$ at machine precision for $q \in \{29, 61, 101, 151\}$	**proved**
Q5b	4D Lorentzian geometry	[9]	BFS stratification of $\mathrm{Heis}_3(\mathbb{R}) \to$ 4D Lorentzian spacetime; conditional	**structural**
Q6a	Gauge structure	[19]	U(1) and SU(2) from fibre fixed points; $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ derived (O31 [63], conditional on [H-color])	**structural**
Q6b	Effective spacetime	[14]	Effective Lorentzian geometry from Π_q; inherits Q5a–Q5b hypotheses	**structural**
Q7	Dimensional bridge	[65]	$H_{\mathrm{eff}} \cong \mathrm{Sym}^2(V_\rho)$; $[F_c, L_{\mathrm{Weil}}] = 0$; bridge non-obstruction (Case 3 excluded by Q9); $A_z = C_{\mathfrak{su}(2)} = 2$; uniqueness of φ up to scalar	**structural**
Q8	Casimir rigidity	[49]	$A_Z = C_{\mathfrak{su}(2)} = 2$ unconditionally under Q5a; λ fixed by horizontal sector consistency; $g^{\mu\nu} = \mathrm{diag}(-A_\tau, 2, 2, 2)$ established	**structural**
Q9	[H-lift] proved	[60]	Kinetic-sector identification: modulation generator suppressed at rate $O(q^{-1/2})$; [H-lift] is a theorem under Q5a; $A_H > 0$ proved by coercivity, independently of bridge	**proved**
Q10	Asymptotic isotropy	[58]	$A_H(q) \to 2$ with $\lvert A_H - 2\rvert \leq C\,\varepsilon(q)$; universality [U] $\Rightarrow$ $\mathfrak{su}(2)$-isotropy $\Rightarrow$ Casimir; bridge conjecture resolved (isotropic case) under [U]	**structural**
U1	Universality [U] proved	[66]	$\varepsilon(q) = O(q^{-1/2})$ from Weil-generator Lipschitz bound and BFS–Carnot–Carathéodory convergence; no BI input required for small-θ regime	**proved**
W1	[H-w] proved	[68]	$a_q(s) \to A > 0$ uniformly in s: fibre-invariance of admissible observables ($\mathcal{O} = \mathrm{Im}\,\Pi$) is the structural origin; U1 controls the rate $O(q^{-1/2})$; after W1 + Q5a-O2 [62], open Q5a hypotheses reduced to [H2] only	**proved**

continued...

Label	Short title	Key	Main contribution	Status
H2	[H2] proved	[61]	Strong convergence $\hat{X}_q \to x$, $\hat{P}_q \to -i\partial_x$ on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$: quantitative Poisson aliasing lemma (rate $O(q^{-1})$), discrete Sobolev identity, quasi-isometry of sinc embedding; closes last open hypothesis of Q5a Theorem 7.1	proved
Q11	Temporal Casimir rigidity	[64]	Cascade internalization: ∂_τ is the Carnot limit of $T = \Delta_n$; [U] forces T SU(2)-equivariant; Schur + Q8 normalisation give $A_\tau = 2$; $g^{\mu\nu} = 2\,\eta^{\mu\nu}$; Q5b-O3 closed	structural
Spec2	Relational spacetime	[42]	Operational metric from graph Laplacians; spacetime from spectral correlations	structural
Gravity	IR Einstein response	[21]	$G_{\mu\nu} = 8\pi G_N T^{(\Pi)}_{\mu\nu}$ from $\delta S_\Pi/\delta g$ at leading IR order	proved
Thermo	Spectral thermodynamics	[23]	Local spectral first law; Einstein equation as spectral equilibrium condition	structural
Lorentz	Gravitational waves	[11]	Lorentzian completion; causal propagation; GW spectrum	proved
TopInv	Topological invariants	[54]	Charge quantisation ($\pi_1 \cong \mathbb{Z}$), no monopoles, chirality (Hopf fibration)	proved
SMcm	Charge, mass, inertia	[12]	Mass and charge as BI-saturated responses	structural
Bell	Bell non-locality	[8]	Bell violations from non-injectivity alone	proved
SchLamb	Fine-structure constant	[20]	α as projective invariant; Lamb shift and Schwinger correction	structural
Cosmol	Rotation curves / Hubble	[45]	Flat rotation curves and Hubble tension from a common effective mechanism	structural
Supra	Superconductivity	[55]	Projective phase locking as origin of superconducting coherence	structural

7 Open Problems

7.1 Branch II (O-Series)

Extension of O29 to larger primes. The symmetric rank formula is confirmed for $q \in \{61, 101, 151, 211\}$ (103/105 pairs at $q = 211$, 98%). Extension of the numerical computation to $q \in \{307, 401\}$ is pending.

Non-conjugate block protocol. A measurement protocol using non-conjugate block data (e.g. the c-block at two independent primes) would probe the full rank $d_\rho^2 = 4$ and provide a direct test of O26 Criterion 5.4 in its original formulation.

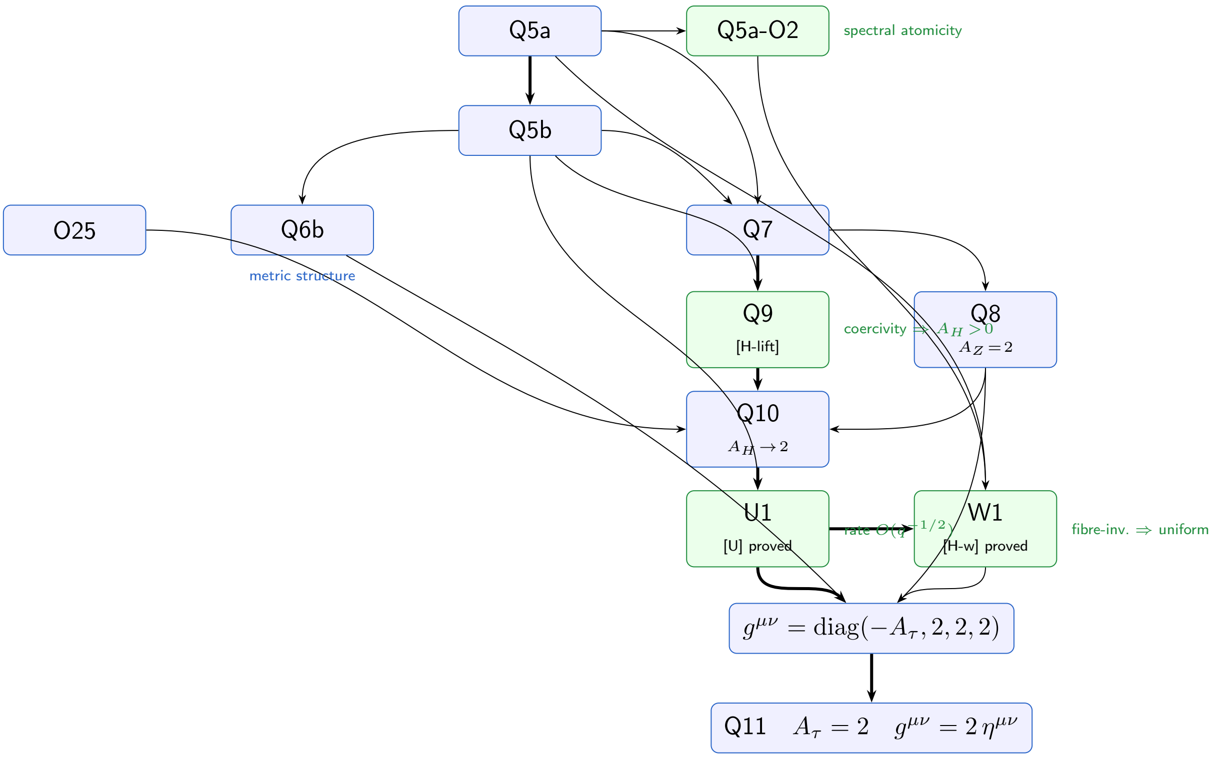

Figure 2: Geometric emergence sub-programme: logical chain from Q5a to the effective Lorentzian co-metric. **Q5a-O2** [62] (**proved**): spectral atomicity of $\mathrm{ran}(\Pi_q)$; proves [H-$\mathcal{E}$1] and [C]; closes all Q5a hypotheses except [H2]. **Q6b** [14] (**structural**): derives the effective Lorentzian metric structure from the admissibility operator L_{eff} of Q5b; closes the link $\Pi_q \xrightarrow{Q5a} L_{\mathrm{eff}} \xrightarrow{Q6b} g^{\mu\nu}$. **Q9** (**proved**): [H-lift] is a theorem under Q5a; coercivity gives $A_H > 0$ independently of the bridge. **Q8** (**structural**): $A_Z = C_{\mathfrak{su}(2)} = 2$ by Casimir rigidity; normalisation fixed by horizontal sector. **Q10** (**structural**): universality [U] ⇒ $\mathfrak{su}(2)$-isotropy ⇒ $A_H \to 2$, conditional on [U]. **U1** (**proved**): [U] proved with rate $O(q^{-1/2})$ from Weil-generator Lipschitz bound and BFS–Carnot–Carathéodory convergence (Q5b). **W1** (**proved**): [H-w] proved as a structural consequence of fibre-invariance ($\mathcal{O} = \mathrm{Im}\,\Pi$), quantitative control from U1. The co-metric $g^{\mu\nu} = \mathrm{diag}(-A_\tau, 2, 2, 2)$ is established under the Q5a hypotheses; [H2] is the sole remaining open hypothesis. **Q11** (**structural**): cascade internalization identifies ∂_τ with the BFS cascade generator via Carnot–Carathéodory convergence; [U] forces SU(2)-equivariance; Schur's lemma and Q8 normalisation give $A_\tau = 2$. The effective metric is uniquely determined: $g^{\mu\nu} = 2\,\eta^{\mu\nu}$. Q5b-O3 is closed.

Analytical control of $n_1(q)/q$. The BFS window ratio $n_1(q)/q$ decreases monotonically through $q = 211$ (O28, Scenario S2); the asymptotic constant $\hat{\alpha}$ is bounded below by 0.062 (updated from O28 with $q = 211$ data) but not yet stabilised.

Notation consolidation: $c_\chi \to c_{\mathrm{BI}}$. A coordinated rename of the BI saturation constant across all papers is planned and deferred pending consistency with already-deposited O-series papers.

7.2 Branch III (Q-Series and Applications)

Bridge existence at finite q (Q7–Q11). The asymptotic isotropy $A_H \to 2 = A_Z$ is proved by Q10 and U1 (conditionally on [U]), and $A_\tau = 2$ is proved by Q11 (conditionally on [U]), establishing $g^{\mu\nu} = 2\,\eta^{\mu\nu}$ and closing Q5b-O3. What remains open is the finite-q structural question: whether an explicit $\mathfrak{su}(2)$-equivariant isomorphism $\varphi_q : \mathrm{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\mathrm{sp}}$ exists at each prime q, not only in the limit (Q11-O1). A complementary derivation of A_τ from Born–Infeld saturation data [38] independently of [U] is also open (Q11-O2).

$\pi_2(\mathcal{C}_{\mathrm{eff}}) = 0$ beyond the canonical model (TopologicalInvariants, Remark 4.2). General proof of this topological property is open.

Full SM gauge group: [H-color] and O32 (Q6a, O31). O31 [63] derives $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ as a composite of admissibility fixed points unconditionally for the colour-adapted Cayley graph, and conditionally on Hypothesis [H-color] for the standard graph. What remains open is the derivation of [H-color] from Foundation M axioms, and its numerical verification in O32 ($q \in \{61, 151, 211, 307\}$, $q \equiv 1 \pmod 3$).

Baryon-to-photon ratio (SMchargemass). Quantitative derivation of $\eta \sim 10^{-10}$ is open.

$V - A$ chiral structure. Formal derivation of the $V - A$ coupling from spectral selection is open.

Thermodynamics: Lorentzian extension and matter coupling. The local spectral first law is derived in the Riemannian setting. Analytic continuation and the explicit form of W_Π for matter fields are open.

8 Conclusion

The Cosmochrony programme has established a coherent, multi-level derivation of physical structure from a single primitive. Branch I (Foundation, ENI, HeisenbergStructure [27]) is formally closed: the Weil representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ is a theorem. The binary-polyhedral maximality conjecture (SpectralCapacity [47]) is proved: $2I$ uniquely maximises $\mathcal{C}_\alpha(G,S)$ for all $\alpha > 0$ among finite SU(2) subgroups with neutral generating sets, closing the precursor programme for the O-series. Branch II (O-series) has closed the unconditional transfer chain $c_\chi \to \delta_{\mathrm{pair}} \to \beta^*$ analytically (O24) and numerically validated for $q \in \{29, 61, 101, 151, 211\}$ (O25). The spin-$\frac{1}{2}$ sector identification is complete (O29): the symmetric rank formula $r_{\mathrm{eff}} = d_\rho(d_\rho+1)/2$ uniquely identifies $d_\rho = 2$ from the observed $r_{\mathrm{eff}} = 3$, confirmed numerically for $q \in \{61, 101, 151\}$. The invariant eigenvalue ratio $\lambda_1/\lambda_2 = 2$ of the per-pair covariance is now analytically derived (O30): the Born–Infeld parity anti-linearity forces an equatorial constraint on the admissible fibre, and the factor of 2 is purely structural—the squared symmetric-tensor normalisation of the Carnot-degree-2 central generator $Z = [X, Y]$. O31 develops the structural framework for SU(3): the metaplectic intertwining of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ forces triplet co-admissibility unconditionally for the colour-adapted Cayley graph, and SU(3) is thereby identified as the symmetry group of the co-admissible colour triplet, conditional on Hypothesis [H-color] for the standard graph. Branch III (Q-series and companions) derives quantum mechanics, spacetime, and SM observables at the structural level, with several results conditional on explicitly stated hypotheses. The SU(2) quantum sector is now complete: Q3 closes the general-j open step of Q2 by identifying the proto-state as the singlet for all five admissible sectors of $2I$ via Schur's lemma, and derives the universal correlator $E(\hat{a}, \hat{b}) = -j(j+1)/3 \cdot (\hat{a} \cdot \hat{b})$ and Born rule for $j \in \{\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}\}$ (**proved**). The geometric emergence sub-programme has advanced substantially: Q9 proves [H-lift] and $A_H > 0$ (unconditionally under Q5a); Q10 proves $A_H \to 2$ under spectral universality [U]; U1 proves [U] with rate $O(q^{-1/2})$ from BFS convergence and Weil-generator stability; W1 proves [H-w] as a structural consequence of the fibre-invariance of admissible observables. The effective Lorentzian co-metric $g^{\mu\nu} = \mathrm{diag}(-A_\tau, 2, 2, 2)$ is thereby established under the Q5a foundational hypotheses and [U]; Q11 [64] closes open problem Q5b-O3 by identifying $A_\tau = 2$ via cascade internalization and Casimir rigidity, yielding the uniquely determined metric $g^{\mu\nu} = 2\,\eta^{\mu\nu}$ with no remaining free parameter. Q6b [14] establishes the middle link of the geometry derivation chain, extracting the effective Lorentzian metric structure directly from the admissibility operator L_{eff} of Q5b and thereby closing the chain $\Pi_q \xrightarrow{Q5a} L_{\mathrm{eff}} \xrightarrow{Q6b} g^{\mu\nu} \xrightarrow{\text{Gravity}} G_{\mu\nu}$ (**structural**, conditional on Q5a). Q5a v2.0 establishes that [H1] is not needed in full generality on the admissible sector. Q5a-O2 [62] proves spectral atomicity of the admissible sector and closes [H-$\mathcal{E}$1] and [C]: each conjugate pair contributes exactly three pure Fourier modes, coercivity at the optimal diffusive scale follows from the non-zero limiting frequency, and Mosco compactness reduces to Bolzano–Weierstrass in $\mathbb{C}^2$. The H2 paper proves the last open hypothesis [H2] (strong convergence of the rescaled Weil generators $\hat{X}_q \to x$ and $\hat{P}_q \to -i\partial_x$) via a quantitative Poisson aliasing lemma and a discrete Sobolev identity; all hypotheses of Q5a Theorem 7.1 are verified on the admissible sector, and the large-q convergence theorem holds unconditionally.

For an external reader, three entry sequences are suggested. For the quantum structure thread: ENI → Foundation → O18 → O22 → O23 → Q1. For the spectral mass hierarchy thread: ENI → Foundation → O16 → O24. For the spacetime thread: Spectral2 → Gravity → (Thermodynamics or Lorentz); for the geometric emergence thread: Q5a → Q5b → Q6b → Q7 → Q9 → Q10 → U1 → W1 → Q11.

References

[1] J. Beau. Three-step path fingerprints on lps graphs and the capacity growth exponent, 2025. Spectral admissibility sub-program "O8" paper, doi:10.5281/zenodo.19151769.

[2] J. Beau. About canonical pair observables in weil blocks: Structure of the residual amplitude factor and pipeline-independent formulation, 2026. O19 paper, doi:10.5281/zenodo.19323720.

[3] J. Beau. Admissible frontier saturation and the cascade exponent: A representation-theoretic obstruction and its matrix-level refinement, 2026. Spectral admissibility sub-program "O5" paper, doi:10.5281/zenodo.19120328.

[4] J. Beau. Admissible non-injective transitions as the primitive of physical description, 2026. Cosmochrony Foundation Paper "M", doi:10.5281/zenodo.19616329.

[5] J. Beau. Asymptotic calibration of the bfs window and effective dimension of the admissible trajectory, 2026. O28 paper, doi:10.5281/zenodo.19767801.

[6] J. Beau. Asymptotic saturation of projective resolution: Expander relaxation graphs, 2026. Spectral admissibility sub-program paper, doi:10.5281/zenodo.19056975.

[7] J. Beau. Asymptotic stability of exact weil-block capacity on heisenberg graphs: Extended prime range, variance reduction, and requalification of the $\delta \to \beta^*$ tension, 2026. Spectral admissibility sub-program "O13" paper, doi:10.5281/zenodo.19210831.

[8] J. Beau. Bell-inequality violations from non-injective projection, 2026. Cosmochrony companion paper B, doi:10.5281/zenodo.18371173.

[9] J. Beau. Bfs shell stratification and the emergence of four-dimensional lorentzian geometry, 2026. spectral admissibility subprogram "Q5b" paper, doi:10.5281/zenodo.19686700.

[10] J. Beau. Bounded relaxation and the dynamical selection of spacetime geometry, 2026. Cosmochrony companion paper C about Born-Infeld, doi:10.5281/zenodo.18407505.

[11] J. Beau. Causal propagation and gravitational waves from projective spectral dynamics, 2026. Cosmochrony companion paper J, doi:10.5281/zenodo.18826644.

[12] J. Beau. Charge, mass, and inertia as saturated responses in a pre-geometric relational description, 2026. Cosmochrony companion paper E, doi:10.5281/zenodo.18488987.

[13] J. Beau. Cosmochrony: A non-injective projection theory of emergent physics, 2026. original white paper, doi:10.5281/zenodo.17957509.

[14] J. Beau. Effective spacetime geometry from admissible non-injective projection, 2026. spectral admissibility subprogram "Q5b" paper, doi:10.5281/zenodo.19744005.

[15] J. Beau. Exact weil-block projective capacity on heisenberg graphs: Resolving the final obstruction to δ extraction, 2026. Spectral admissibility sub-program "O12" paper, doi:10.5281/zenodo.19194798.

[16] J. Beau. Fibre admissibility and resolution of the capacity exponent discrepancy, 2026. Spectral admissibility sub-program "O16" paper, doi:10.5281/zenodo.19298427.

[17] J. Beau. Fibre-level admissibility from conjugate weil blocks: Structural derivation of the pair observable, 2026. Cosmochrony Spectral Admissibility sub-Program "O17" paper, doi:10.5281/zenodo.19299611.

[18] J. Beau. From admissibility to quantum structure: Phase coherence and correlations from non-injective projection, 2026. spectral admissibility subprogram "Q1" paper, doi:10.5281/zenodo.19561060.

[19] J. Beau. Gauge structure from admissible non-injective projection, 2026. spectral admissibility subprogram "Q5a" paper, doi:10.5281/zenodo.19655294.

[20] J. Beau. he fine-structure constant as a projective invariant: Lamb shift and schwinger effect in a pre-geometric framework, 2026. Cosmochrony companion paper F, doi:10.5281/zenodo.18602702.

[21] J. Beau. Infrared einstein response from a renormalized spectral entropy functional, 2026. Cosmochrony companion paper H, doi:10.5281/zenodo.18818721.

[22] J. Beau. Large-q limit of the admissible fibre: Representational and operator convergence, 2026. spectral admissibility subprogram "Q5a" paper, doi:10.5281/zenodo.19642369.

[23] J. Beau. Local spectral thermodynamics and the einstein equation as a spectral equilibrium condition, 2026. doi:10.5281/zenodo.18825655.

[24] J. Beau. Matrix-level dynamic redundancy and the structural confinement of the cascade exponent β, 2026. Spectral admissibility sub-program "O6" paper, doi:10.5281/zenodo.19140624.

[25] J. Beau. Minimal fibre structure of the non-injective projection from born–infeld indiscernability: Derivation of the parity involution, 2026. Spectral admissibility sub-program "O18" paper, doi:10.5281/zenodo.19321413.

[26] J. Beau. No external dimensional scale at the level of admissibility, 2026. Cosmochrony structural note, doi:10.5281/zenodo.19744767.

[27] J. Beau. Non-factorisability and the emergence of heisenberg structure from admissibility constraints, 2026. spectral admissibility subprogram paper, doi:10.5281/zenodo.19635395.

[28] J. Beau. Non-injectivity as a structural necessity of genuine emergence: A no-go theorem for faithful emergent descriptions, 2026. Cosmochrony companion paper "ENI" or L, doi:10.5281/zenodo.19391746.

[29] J. Beau. Observable-class derivation mismatch and the exact growth equation: From proxy capacity to block-level dynamics, 2026. O15 spectral admissibility subprogram paper, doi:10.5281/zenodo.19281420.

[30] J. Beau. Observable-class mismatch and the corrected $\delta \mapsto \beta^*$ relation on heisenberg graphs: Block normalisation, central-phase contribution, 2026. O14 spectral admissibility subprogram paper, doi:10.5281/zenodo.19226272.

[31] J. Beau. Pairwise character geometry and the classification of neutral generators in su(2), 2026. Spectral admissibility sub-program "rigidity" paper, doi:10.5281/zenodo.18969867.

[32] J. Beau. Polynomial-growth capacity dynamics at large q: Volumetric growth confirmation, capacity decay, and the weil decomposition roadmap, 2026. Spectral admissibility sub-program "O10" paper, doi:10.5281/zenodo.19161074.

[33] J. Beau. Projective capacity and the continuum limit of admissible redundancy, 2026. Spectral admissibility sub-program "O7" paper, doi:10.5281/zenodo.19149773.

[34] J. Beau. Projective capacity beyond expanders: Polynomial-growth relaxation graphs and the capacity exponent, 2026. Spectral admissibility sub-program "O9" paper, doi:10.5281/zenodo.19154517.

[35] J. Beau. Projective dynamics and mass hierarchy: Hierarchical amplification via growing relational valence, 2026. Spectral admissibility sub-program "O3" paper, doi:10.5281/zenodo.19076244.

[36] J. Beau. Projective persistence and the physical sub-spectrum: A dynamical selection criterion for the capacity exponent, 2026. Spectral admissibility sub-program "O20" paper, doi:10.5281/zenodo.19339888.

[37] J. Beau. Projective resolution dynamics on variable-valence relaxation graphs, 2026. O1 spectral admissibility subprogram "O1" paper, doi:10.5281/zenodo.19076319.

[38] J. Beau. Projective temporal ordering and cumulative projected capacity: Numerical evidence from the spectral admissibility cascade, 2026. Spectral Admissibility sub-Program temporal ordering paper, doi:10.5281/zenodo.19474639.

[39] J. Beau. Quadratic completion of admissible spectral pairs via binary-icosahedral representation, 2026. O26 paper, doi:10.5281/zenodo.19488845.

[40] J. Beau. Quantum structure beyond spin-1/2: Co-admissible sectors and the emergence of su(2) as a fixed point authors/creators, 2026. spectral admissibility subprogram "Q2" paper, doi:10.5281/zenodo.19616444.

[41] J. Beau. Quaternionic rigidity of admissible morphisms: Every admissible $\phi_{q,\rho}$ necessarily factors through $\mathfrak{su}(2)$, 2026. O27 paper, doi:10.5281/zenodo.19489296.

[42] J. Beau. Relational reconstruction of spacetime geometry from graph laplacians, 2026. Cosmochrony companion paper A, doi:10.5281/zenodo.18356037.

[43] J. Beau. Shell-alignment from projection locking: A discrete admissibility theorem under born-infeld saturation, 2026. Spectral admissibility sub-program "O22" paper, doi:10.5281/zenodo.19367375.

[44] J. Beau. Shell-level saturation and the fibre-level transfer $\delta_{\text{pair}} \to \beta^*$, 2026. Spectral admissibility sub-program "O21" paper, doi:10.5281/zenodo.19358961.

[45] J. Beau. A single effective framework underlying galaxy rotation curves and the hubble tension, 2026. Cosmochrony companion cosmology paper D, doi:10.5281/zenodo.18452299.

[46] J. Beau. Spectral admissibility under bounded flux: Non-abelian mode selection on cayley graphs of su(2) subgroups, 2026. Step A of Cosmochrony Spectral Admissibility sub-Program, doi:10.5281/zenodo.18944985.

[47] J. Beau. Spectral capacity functional and the binary-polyhedral maximality conjecture, 2026. Step B of Cosmochrony Spectral Admissibility sub-Program, doi:10.5281/zenodo.18945273.

[48] J. Beau. Spin-$\frac{1}{2}$ sector identification via the symmetric rank formula: Effective dimension of the admissible covariance in $\mathrm{End}(v_\rho)$, 2026. O29 paper, doi:10.5281/zenodo.19787221.

[49] J. Beau. Sub-principal symbol of the effective operator and casimir rigidity of the central direction, 2026. "Q8" paper, doi:10.5281/zenodo.19879909.

[50] J. Beau. Subdiffusive valence growth under bounded relational flux: Structural derivation of the cascade exponent β, 2026. Spectral admissibility sub-program "O4" paper, doi:10.5281/zenodo.19101472.

[51] J. Beau. Systematic pair-level campaign for δ_{pair}: Convergence, inter-pair concentration, and normalization structure, 2026. O25 paper, doi:10.5281/zenodo.19462132.

[52] J. Beau. Three particle generations and mass hierarchy from spectral stratigraphy, 2026. Spectral admissibility sub-program paper, doi:10.5281/zenodo.19040490.

[53] J. Beau. Three stable directions from quaternionic minimality: Derivation of $\sigma_c(n_3) = 3$ from born–infeld fibre admissibility, 2026. Spectral admissibility sub-program "O23" paper, doi:10.5281/zenodo.19375136.

[54] J. Beau. Topological invariants of admissible configurations: Charge quantisation, absence of magnetic monopoles, and projective chirality in cosmochrony, 2026. Cosmochrony companion paper K, doi:10.5281/zenodo.19281420.

[55] J. Beau. A unified real-space frustration mechanism for superconductivity: Projective phase locking from conventional to strongly correlated systems, 2026. Cosmochrony companion paper G, doi:10.5281/zenodo.18718224.

[56] J. Beau. Vertical non-injectivity and the stability of the observable rank: Closing the unconditional transfer $c_\chi \to \delta_{\text{pair}} \to \beta^*$, 2026. O24 spectral admissibility subprogram "O24" paper, doi:10.5281/zenodo.19386581.

[57] J. Beau. Weil-block projective capacity on heisenberg graphs: Resolving the representation obstruction and extracting the capacity exponent, 2026. Spectral admissibility sub-program "O11" paper, doi:10.5281/zenodo.19169987.

[58] Jérôme Beau. Asymptotic $\mathfrak{su}(2)$-isotropy of the effective quadratic form and the identification $a_h = 2$, 2026. "Q10" paper, doi:10.5281/zenodo.19880900.

[59] Jérôme Beau. Born–infeld parity forces an equatorial admissible fibre: Analytical derivation of the $[1 : \frac{1}{2} : \frac{1}{2}]$ eigenvalue ratio in $\mathrm{Sym}^2(v_\rho)$, 2026. "O30" paper, doi:10.5281/zenodo.20014707.

[60] Jérôme Beau. Proof of the lifting hypothesis [H-lift]: Kinetic-sector identification via generator suppression, 2026. "Q9" paper, doi:10.5281/zenodo.19880574.

[61] Jérôme Beau. Semiclassical consistency of the weil representation on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$: Quantitative sinc embedding, generator convergence, and aliasing control, 2026. "H2" paper, doi:10.5281/zenodo.19962051.

[62] Jérôme Beau. Spectral atomicity of the admissible sector: Scaled coercivity and mosco compactness without nash inequalities, 2026. "Q5a-O2" paper, doi:10.5281/zenodo.19915951.

[63] Jérôme Beau. Su(3) from colour triplet co-admissibility: Towards the complete standard model gauge group, 2026. "O31" paper, doi:10.5281/zenodo.20030491.

[64] Jérôme Beau. Temporal casimir rigidity and closure of the effective metric: Identification of $a_\tau = 2$, 2026. "Q11" paper, doi:10.5281/zenodo.20098387.

[65] Jérôme Beau. Three admissible directions and three spatial dimensions: A structural bridge candidate, 2026. "Q7" paper, doi:10.5281/zenodo.19802123.

[66] Jérôme Beau. Uniform spectral universality for weil fingerprint energies: proof of [U] with rate $o(q^{-1/2})$, 2026. "U1" paper, doi:10.5281/zenodo.19881146.

[67] Jérôme Beau. Universal spin-j singlet from admissibility: Completion of the SU(2) quantum sector, 2026. "Q3" paper, doi:10.5281/zenodo.19979535.

[68] Jérôme Beau. Weight stabilisation of the admissible dirichlet form: Proof of hypothesis [h-w] from spectral universality, 2026. "W1" paper, doi:10.5281/zenodo.19886319.